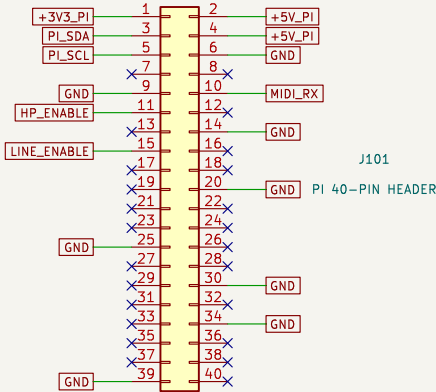


ARDOR / CODEC ZERO I/O

01 / System interface and power

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

PI HEADER / STACKING INTERFACE

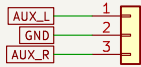


GPIO15 / pin 10: UART RX, 31250 baud, 8–N–1
GPIO17 / pin 11: headphone enable, active high
GPIO27 / pin 13: reserved for Codec Zero button

GPIO22 / pin 15: line relay enable, active high
I2C1: GPIO2/3, ADS1115 address 0x48.
Reserved Codec pins: GPIO18–21 I2S; GPIO23/24 LEDs;
GPIO27 button; GPIO0/1 HAT EEPROM. No added loads.

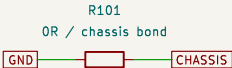
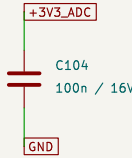
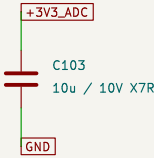
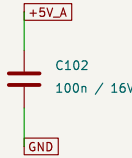
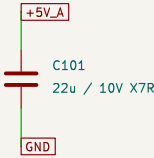
AUX HEADER HARNESS

J102
CODEC AUX L / GND / R



J102 pin order is OUR harness definition.
Map to the actual Codec Zero L/R/GND pads.
Never connect the mono speaker BTL output here.

FILTERED BRANCHES / 5 V REGULATED INPUT ONLY



Bond enclosure at connector entry. Short, wide copper.
Use one continuous PCB ground plane; place by function.
Do not route ESD or charge-pump return through ADC/audio.
5 V source: 4.75–5.25 V; reserve 150 mA beyond Pi/Codec.
Pi provides 3.3 V. No second supply or USB back-feed.

TP101
+5V_A



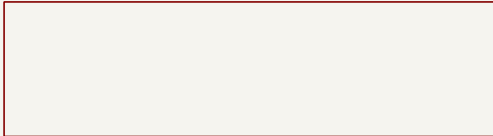
TP102
+3V3_ADC



TP103
GND

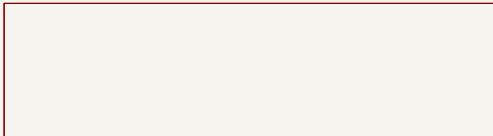


02 / Isolated MIDI input



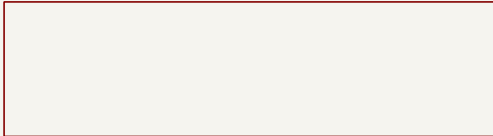
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03 / Expression input



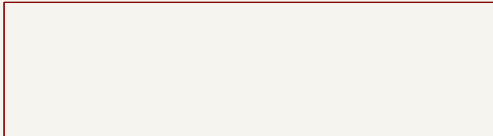
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04 / Audio distribution



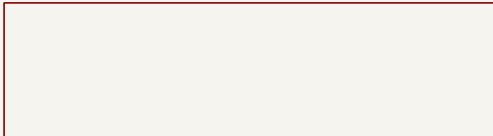
File: 04_Audio.kicad_sch

05 / Line and amp feed



File: 05_Line_Amp.kicad_sch

06 / Stereo headphones



File: 06_Headphones.kicad_sch

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /
File: Ardor_I0.kicad_sch

Title: 01 / System interface and power

Size: A3

Date: 2026–09–05

Rev: A

KiCad E.D.A. 9.0.2

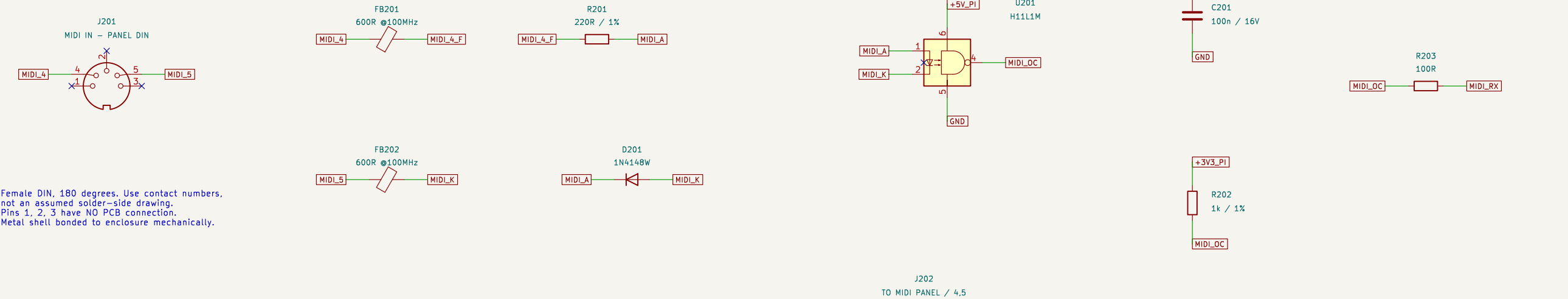
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ARDOR / CODEC ZERO I/O

02 / Isolated MIDI input

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

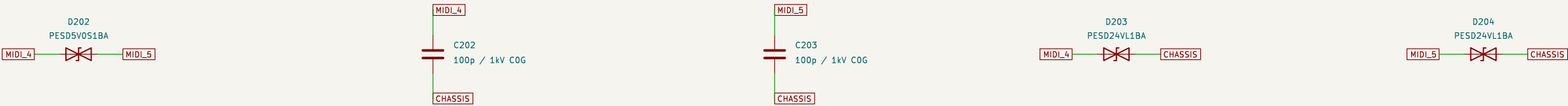
5–PIN DIN / FLOATING CURRENT LOOP



Female DIN, 180 degrees. Use contact numbers, not an assumed solder-side drawing. Pins 1, 2, 3 have NO PCB connection. Metal shell bonded to enclosure mechanically.

MIDI LED on -> UART low. No inverter required. Open collector pulled up ONLY to 3.3 V. Disable serial console; enable stable UART RX. Never use a 5 V pull-up on Raspberry Pi GPIO.

CONNECTOR ESD / RF NETWORK



D202 clamps differential transients; D203/D204 divert common-mode ESD to CHASSIS (24 V standoff). Keep the MIDI loop copper isolated from logic GND: >= 3 mm clearance around U201 input and nets. Common-mode range is limited by D203/D204. Functional ground-loop isolation; NOT safety isolation. No MIDI-pin connection to logic supply. Place TVS and 100p RF capacitors at DIN harness entry. Qualify IEC 61000-4-2 on the assembled enclosure, including common-mode strikes and UART recovery.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /02 / Isolated MIDI input/
File: 02_MIDI.kicad_sch

Title: 02 / Isolated MIDI input

Size: A3
KiCad E.D.A. 9.0.2

Date: 2026-09-05

Rev: A

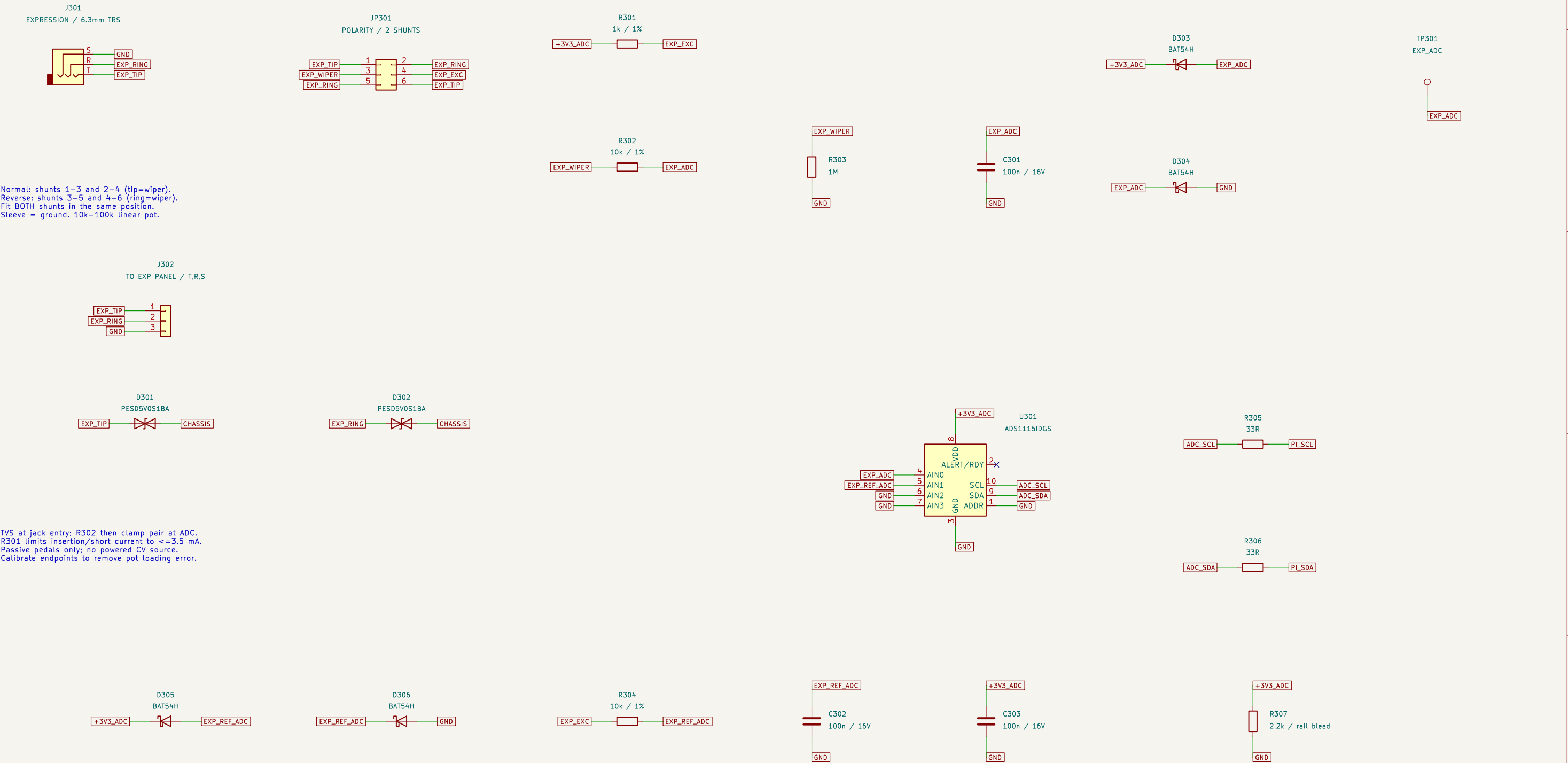
Id: 2/6

ARDOR / CODEC ZERO I/O

03 / Expression input

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

PASSIVE TRS PEDAL / TIP–RING SELECT



Normal: shunts 1–3 and 2–4 (tip=wiper).
Reverse: shunts 3–5 and 4–6 (ring=wiper).
Fit BOTH shunts in the same position.
Sleeve = ground. 10k–100k linear pot.

TVS at jack entry: R302 then clamp pair at ADC.
R301 limits insertion/short current to <=3.5 mA.
Passive pedals only: no powered CV source.
Calibrate endpoints to remove pot loading error.

I2C address 0x48; internal reference; PGA +/-4.096 V; 128 SPS. Read AIN0 and AIN1 single-ended.
Use AIN0 / AIN1 ratio, endpoint calibration, smoothing and deadband. No added I2C pull-ups (Pi has them).

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /03 / Expression input/

File: 03_Expression.kicad_sch

Title: 03 / Expression input

Size: A3

Date: 2026–09–05

Rev: A

KiCad E.D.A. 9.0.2

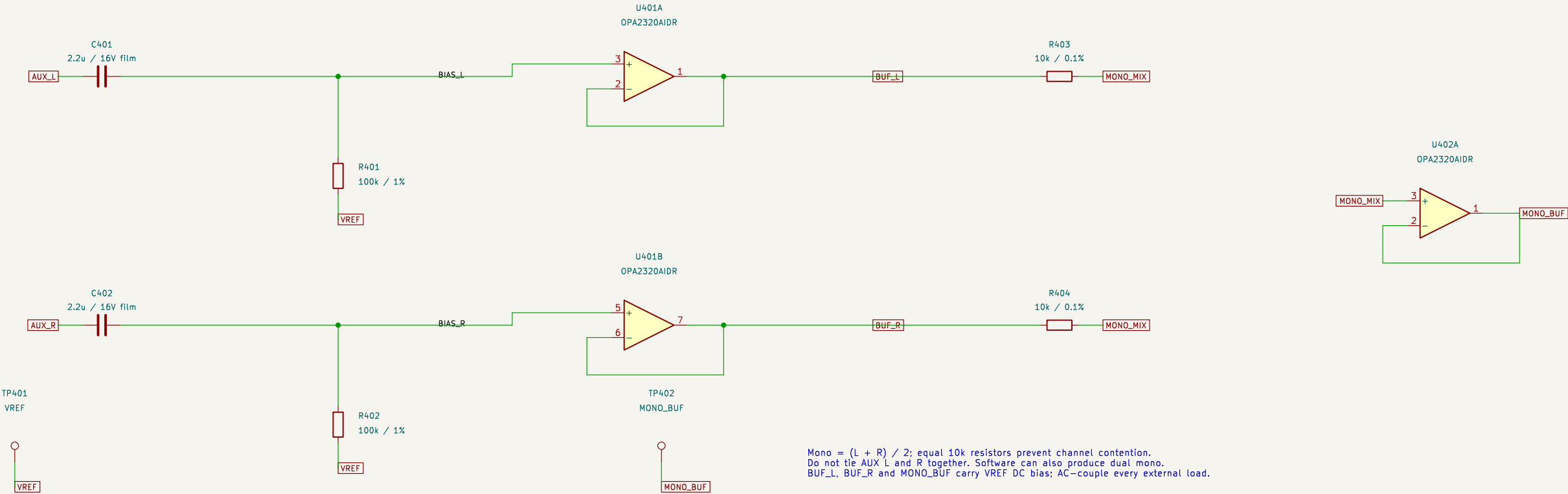
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ARDOR / CODEC ZERO I/O

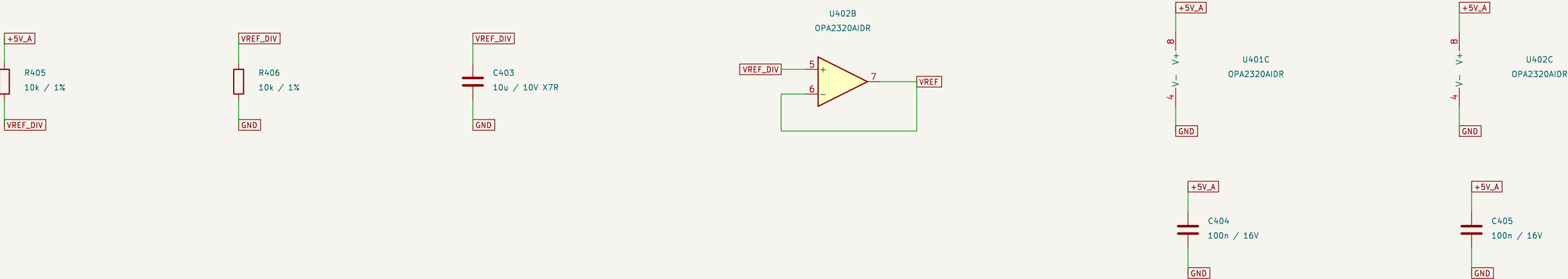
04 / Audio distribution

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

AC-COUPLED STEREO INPUT / UNITY BUFFERS



MIDRAIL / LOCAL BYPASS



Nominal audio target <=1.0 Vrms per AUX channel. Unity gain throughout.
OPA2320: RRIO on 5 V, unity stable; 2.5 V bias permits bipolar audio swing.
2.2u/100k input pole ~0.72 Hz. Keep VREF local; no large capacitor directly on buffer output.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /04 / Audio distribution/
File: 04_Audio.kicad_sch

Title: 04 / Audio distribution

Size: A3
KiCad E.D.A. 9.0.2

Date: 2026-09-05

Rev: A

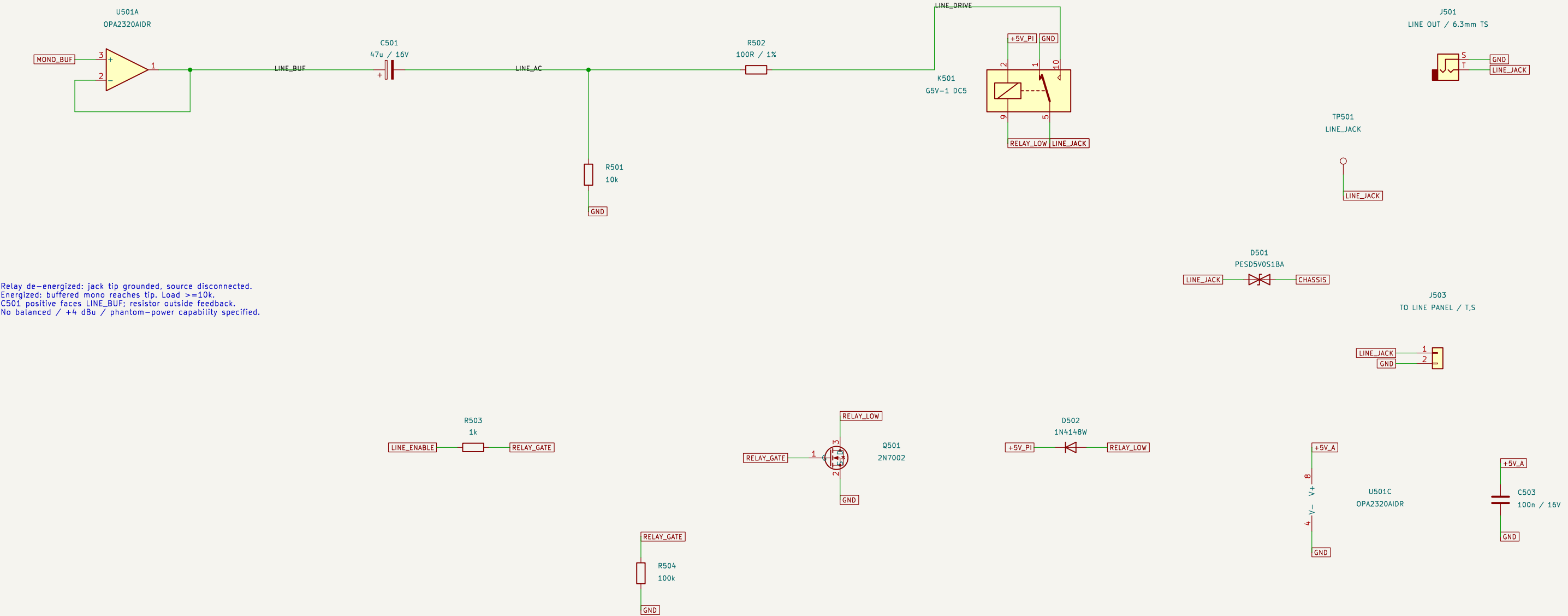
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ARDOR / CODEC ZERO I/O

05 / Line and amp feed

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

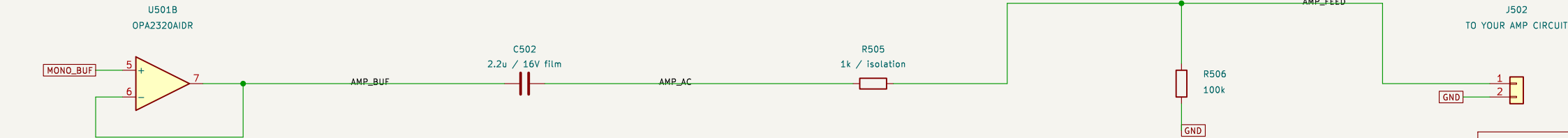
MONO LINE / DC BLOCK / STARTUP RELAY MUTE



Relay de-energized: jack tip grounded, source disconnected.
Energized: buffered mono reaches tip. Load >=10k.
C501 positive faces LINE_BUF; resistor outside feedback.
No balanced / +4 dBu / phantom-power capability specified.

Amp input >=100k. Add your gain, output connector,
external-port protection and mute in that project.
J502 is internal, NOT a speaker / amp output jack.

AMP CIRCUIT PLACEHOLDER / INTERNAL HARNESS



Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /05 / Line and amp feed/

File: 05_Line_Amp.kicad_sch

Title: 05 / Line and amp feed

Size: A3

Date: 2026-09-05

Rev: A

KiCad E.D.A. 9.0.2

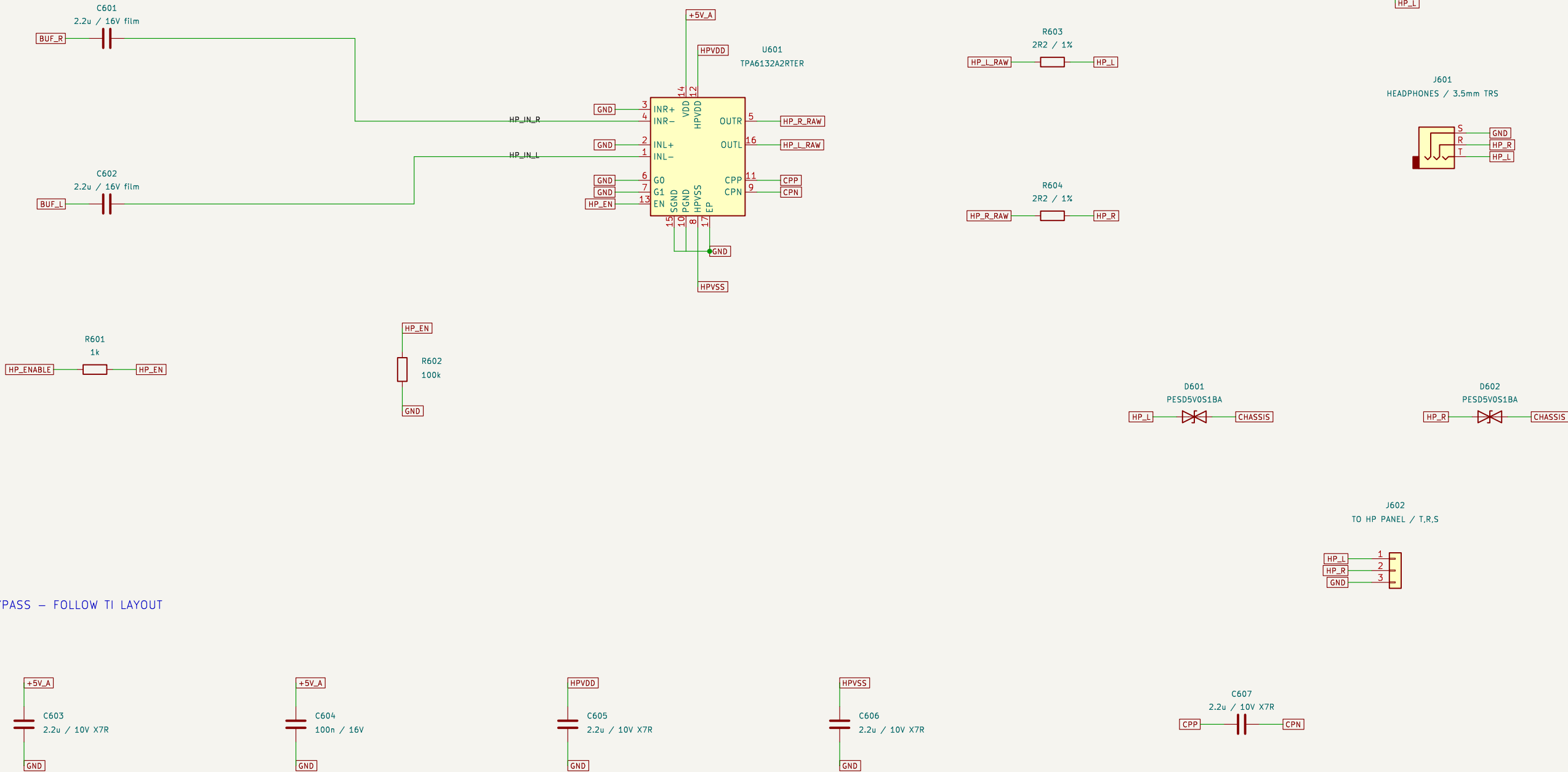
Id: 5/6

ARDOR / CODEC ZERO I/O

06 / Stereo headphones

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

STEREO DIRECTPATH HEADPHONE DRIVER / –6 dB



CHARGE PUMP / BYPASS – FOLLOW TI LAYOUT

HPVDD is an INTERNAL regulator output. Never connect it to +5V_A. HPVSS is negative; use nonpolar ceramic. G0=0, G1=0 gives –6 dB inverting gain. INL+ / INR+ grounded per TI single-ended application. 32 ohm or higher stereo headphones. 1 Vrms AUX gives –0.47 Vrms at 32 ohm (~6.8 mW); start volume low. Keep EN low until codec routing is stable; ramp volume. Short, separate SGND jack return to local ground plane. Place pump capacitors at pins; keep switching loops away from AUX, VREF and expression input traces.

Prototype design; see DESIGN_NOTES.md for release checks

ARDOR – Codec Zero expansion

Sheet: /06 / Stereo headphones/

File: 06_Headphones.kicad_sch

Title: 06 / Stereo headphones

Size: A3

Date: 2026–09–05

Rev: A

KiCad E.D.A. 9.0.2

Id: 6/6